JINMING NIAN

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SUMMARY

Third-year PhD candidate looking for research internship. Specializes in Information Retrieval, Recommendation Systems, NLP, Deep Learning, and LLMs. Experienced in Python and PyTorch, with additional expertise in software development, cloud computing, and distributed systems. Actively publishing research in top-tier conferences on retrieval and LLM-driven methods.

EDUCATION

Santa Clara University *Santa Clara, CA*

Ph.D. in Computer Science and Engineering *September 2023 – June 2028 (Expected)*

Advisor: **Prof. Yi Fang**

M.S. in Computer Science and Engineering *September 2021 – June 2023*

Relevant Courses: Machine Learning, Deep Learning, Information Retrieval, Natural Language Processing, Reinforcement Learning

University of California, Davis *Davis, CA*

B.S. in Physics, Minor in Music *September 2017 – June 2021*

WORK EXPERIENCE

Research Assistant, Santa Clara University *April 2023 - Present*

• Collaboration with Walmart Labs on search and ranking, with recent work on automatic prompt optimization and AlphaEvolve-inspired LLM-as-optimizer methods to improve lexical retrieval algorithms.

Teaching Assistant, Santa Clara University *September 2025 - Present*

• CSEN 169: Web Search & Information Retrieval (designed 4 out of 5 labs in the course), CSEN 166: AI

Data Scientist Intern, Amazon *June 2025 - September 2025*

• Fine-tuned GPT-2 on sub-terabyte scale historical tabular shipment data to forecast Amazon fulfillment center shipment attributes (SLAM time, promise, and volume) during high-volume events such as Prime Day and Black Friday

• Improved SLAM-to-promise time prediction, reducing WAPE from 48.49% to 17.51% at a major Seattle fulfillment center

• Validated the approach's robustness and scalability across multiple fulfillment centers and temporal settings

PUBLICATIONS

[**SIGIR**] **Jinming Nian**, Fangchen Li, Dae Hoon Park, Yi Fang. *RankEvolve: Automating the Discovery of Retrieval Algorithms via LLM-Driven Evolution.* Recently accepted to SIGIR 2026. paper, code

[**SIGIR - ICTIR**] **Jinming Nian**, Zhiyuan Peng, Qifan Wang, Yi Fang. *W-RAG: Weakly Supervised Dense Retrieval in RAG for Open-domain Question Answering.* In Proceedings of the ACM SIGIR International Conference on Innovative Concepts and Theories in Information Retrieval, Padova, Italy, 2025. paper, code

[**SIGIR**] Zhiyuan Peng*, **Jinming Nian***, Alexandre Evfimievski, Yi Fang. *ELOQ: Resources for Enhancing LLM Detection of Out-of-Scope Questions.* In Proceedings of the 48th International ACM SIGIR Conference on Research and Development in Information Retrieval, Padova, Italy, 2025. paper, code, dataset

[**EMNLP Findings**] Xuyang Wu*, **Jinming Nian***, Ting-Ruen Wei*, Zhiqiang Tao, Hsin-Tai Wu, Yi Fang. *Does Reasoning Introduce Bias? A Study of Social Bias Evaluation and Mitigation in LLM Reasoning.* In Findings of the Association for Computational Linguistics: EMNLP, Suzhou, China, 2025. paper

WORK IN PROGRESS

[**Under Review**] **Jinming Nian**, Zhiyuan Peng, Hongwei Shang, Dae Hoon Park, Yi Fang. *Submodular Evaluation Subset Selection in Automatic Prompt Optimization.* 2026. paper, code